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Does the operating temperature of accelerometers affect data measurement during static and dynamic conditions in sports? A randomized-control trial

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Abstract

This study aimed to analyse the effect of two operating temperatures (optimal vs environmental) on IMU internal temperature and accelerometry-based data in two conditions of measurement: (a) static, no movement; and (b) dynamic, a U'16 game-based training soccer session with 14 players. Each session took a 60-min duration and followed an ecological design (no intervention in environmental conditions). Fifty-six WIMU PRO^{TM} inertial devices were randomly grouped in four conditions (two operating temperatures × two measurement conditions, 14 devices in each group) and recorded IMUs' internal temperature, resultant acceleration vector and PlayerLoad (PL_{RT}). Devices were placed in the manufacturer's docking station at the static condition and in a specific harness at the interscapular level at the dynamic condition. Repeated measures *t*-tests were employed for between-group comparisons at 1-min average sections throughout the register. Between-group differences were found in the static and dynamic conditions in IMUs internal temperature (static: 0–31 min; dynamic: 0–20 min) and resultant acceleration vector (static: 0–29 min; dynamic: 0–19 min). Instead, no differences were found in PL_{RT} dynamics and accumulated PL_{RT} although values in the environmental group were overestimated. In conclusion, WIMU PRO^{TM} devices must achieve the optimal operating temperature (38–39°) to avoid measurement error, regardless of the application (short or long sessions).

Keywords

Accelerometers, temperature, workload index, team sports, microtechnology

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Introduction

The use of microsensors technology is a hot topic in sports science due to the large amount of information that can be provided for load quantification and injury prevention out of the lab and in real-world conditions.[1,2] Accelerometers and gyroscopes are the most used microsensors for external workload monitoring in individual and team sports at different body locations, being the most used scapulae and centre of mass attached to the human body by harness or belts respectively.[3,4] In team sports, numerous accelerometry-based indexes (e.g. Player Load, Body Load, Impulse Load, Force Load or Total Acceleration among others) have been developed to quantify external workload in the three planes of movement, finding high-effect size differences that do not allow comparing indexes.[5]

In practical use, the data obtained through accelerometers could be influenced by different factors such as

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sampling frequency, device location, the fixation methods of the devices, the device model and the accelerometry-based workload index, as well as technical aspects that include filtering methods and operating range.[3,4] Sampling frequency determines the amount of data per second and should be at least twice that of the highest magnitude contained in the movement following the Nyquist sampling theorem. Previous research identifies acceptable reliability from 100 Hz to quantify accumulated external workload in sports at the inter-scapulae level.[6,7] Regarding the device location, it should be the closest to the segment that performs the movement/action to achieve the highest accuracy, due to the soft tissues of the human body attenuating the acceleration signal as it travels up through the body.[8,9] To quantify accumulated external workload in sports, a consensus of inter-scapulae location has been reached to achieve the highest tracking accuracy with the accelerometry-based load.[3,4]

The fixation methods of the devices are important to consider because better accuracy and reliability of the measurements were found with greater movement limitation between body and device.[10] The no-movement of the device and the closest distance from the body are achieved by avoiding the use of elastic belts or tape and not placing them on wobbling soft tissues or close to joints.[3] The device model represents another fundamental aspect that influences the data and the magnitude of the measurements due to the different accelerometry-based workload index, as well as their specific filtering methods or operating range.[5,10,11] These aspects limit the comparison of data between devices and between magnitudes of accelerometric load.[3]

Another factor influencing the data collected through accelerometry is the sensor's temperature.[12,13] In this sense, previous studies have shown that three-axis accelerometers usually need a constant temperature using the thermal feedback principle.[14,15] This principle indicates that temperature drift could modify the accelerometers' data, especially when the accelerometers are colder than the optimal operating temperature.[12,13] Thus, due to the lack of research about the effect of internal temperature on accelerometry-based data during practical situations in the sports context, the purpose of this study was to analyse the impact of operating temperature on accelerometry-based data obtained through IMU. We have hypothesized that the operating inertial device temperature directly affects accelerometers' data and therefore the external load indexes based on these sensors' data.

Methods

Design

Fifty-six inertial devices were randomized into four conditions to analyse the effect of two operating temperatures (optimal vs environmental) on IMU internal temperature and accelerometry-based data through resultant vector acceleration in two conditions of measurement (static vs dynamic). Two operating temperature conditions were tested: (a) optimal operating temperature, devices were switched on before the sessions to reach the constant and optimal operating temperature recommended by the manufacturer (38–39°C) and (b) environmental temperature, devices were switched on when the session was going to start. Inertial device temperature was monitored by the internal barometer and environmental temperature by special equipment. For the study purpose, two 60-min trials were designed to evaluate the effect of operating temperature: (a) static trial, 28 devices (14 temperature A condition and 14 temperature B condition) were placed in the manufacturer's docking station to ensure no movement during the measurement, and (b) dynamic trial: 14 U'16 soccer players wore two devices (Temperature A and B condition) at the inter-scapulae line with a specific harness during a game-based training session. Both trials followed an ecological design because environmental conditions were not modified and were realized in the habitual training and competition pitch. All protocols were conducted according to the Declaration of Helsinki and were approved by the Bioethics Commission of the University of Extremadura (register number: 232/2019).

Participants

Fourteen U-16 male soccer players participated voluntarily in the present study (age: 15.8 ± 0.8 years, body mass: 65.1 ± 4.5 kg; height: 175.1 ± 9.7 cm). All players were members of a national-level soccer team. During the study, all players presented an absence of physical limitations and musculoskeletal injuries. Before the session, participants were informed of the risks and discomforts associated with testing and provided written informed consent through their legal guardians.

Equipment

The IMUs internal temperature, resultant vector of acceleration and PlayerLoad were registered by WIMU PRO™ devices (RealTrack Systems, Almeria, Spain) that were composed of four 3D accelerometers with different full-scale output ranges (±16, ±16, ±32, and ±400*g*) for two reasons: (a) increase the reliability through the fusion of the data from the four accelerometers based on the redundancy principle (down-sampling-method),[16] and (b) increase the accuracy of measurements in different output ranges depending on the intensity of the movements.[4] The sampling frequency of microelectromechanical sensors is adjustable from 10 to 1000 Hz, but 100 Hz was set in the present study following a previous study that evaluates this parameter to quantify accumulated external load in sports at the inter-scapulae level.[6] The validity and reliability of accelerometer-based data of WIMU PRO™ have been assessed previously with satisfactory results in different body locations (ankle, knee, lower and upper back).[8,17]

Therefore, considering sampling frequency and fusion of the data from the four accelerometers, one data of each axis (x, y and z) every 0.01 s was extracted to calculate accelerometry-based variables. In addition, this device contained other inertial and tracking sensors (three 3D gyroscopes with 8000°/s full-scale output range, a 3D magnetometer, a barometer, GNSS, UWB, etc.), that detect and measure movement.

In addition, different filtration processes were fulfilled to obtain clean accelerometry-based signals at three levels: (1) accelerometer manufacturer, (2) inertial device chipset and (3) software filtration. These processes are specific to the different sampling frequencies and output ranges of the accelerometers. Users do not control these filtering processes as well as cut-off frequencies used before data are available for the user. The barometer was used for thermal feedback but did not affect accelerometer data recording. Furthermore, each device has an internal 1 GHz microprocessor, temperature sensor, 8 GB flash memories and a highspeed USB interface to record, store and upload data.

Variables

- *Resultant vector of acceleration*: The resultant vector of acceleration is the basis of all formulas used to calculate accumulated workload indexes based on accelerometry in sports.[4] It was calculated as the root square of the sum of the accelerations in the three axes of movement (x, y and z-axis) from the equation (1):

$$\sqrt{(x^2 + y^2 + z^2)} \tag{1}$$

- *PlayerLoad by RealTrack Systems (PL_{RT})*: It is calculated as the vector sum of the Body accelerometric channel calculated through the sensory fusion of inertial device sensors (accelerometer, gyroscope, magnetometer) in its three axes (the vertical, anteroposterior and lateral). PL_{RT} in the current moment was calculated as shown in equation (1) where: PL_n is the player load calculated in the current instant; n is the current instant in time; n–1 is the previous instant in time; X_n, Y_n and Z_n are the values of Body Load for each axis of movement in the current moment in time; X_{n-1}, Y_{n-1} and Z_{n-1} are the values of Body Load for each axis of motion in the last moment in time.[4] Then, the sum of PL_n during the session is calculated and multiplied by 0.01 as a scale factor following the equation (3). It is represented in arbitrary units (a.u.).

$$PL_n = \sqrt{\frac{(X_n - X_{n-1})^2 + (Y_n - Y_{n-1})^2 + (Z_n - Z_{n-1})^2}{100}} \tag{2}$$

$$Accumulated\ PL_{RT} = \sum_{n=0}^{m} PL_n \times 0.01 \tag{3}$$

- *Internal temperature*: It was measured by the barometer included in the WIMU PRO™ device and it was represented in Celsius grades.

Procedures

The 58 inertial devices that were used in the research were randomized into four groups: (1) static condition and environmental temperature, (2) dynamic condition and environmental temperature, (3) static condition and optimal operating temperature and (4) dynamic condition and optimal operating temperature. Each group was composed of 14 inertial devices. Then, the devices of optimal operating temperature groups (3 and 4) were switched on. From the devices' switch on, the operating inertial device temperature was monitored through the barometer of the WIMU PRO™ inertial device second-by-second and visualized in the software SPRO™. Finally, when the inertial devices achieved a constant temperature for 1 min (variations < 0.1°C), it was considered stabilized in optimal working temperature. When the optimal working temperature group achieved the constant temperature, two trials were performed under different conditions:

- *Static trial*. All devices were recorded for 60 min in the manufacturer's docking station to ensure no movement during the measurement.
- *Dynamic trial*. A 60-min game-based training session of a U'16 national-level soccer match was registered. Before the session started, the devices were placed on the players. Each player wore two inertial devices, one of the optimal operating temperature group and one of the environmental temperature group at the interscapular line, through a specific harness (Figure 1). The device pairs were placed randomly in parallel (left-right) with a 2-cm gap between devices at the same height. Then, the training session started based on technical-tactical aspects through different small-sided games. At the end of the training session, the devices were removed from the soccer players and the data were downloaded to a computer for further analysis.

During the study, the environmental temperature was measured through specific equipment (QuestTemp 36, 3M, Minnesota, USA). The temperature (in Celsius grades) at the start was 27.6° in the static trial and 27.7° in the dynamic trial, and the average temperature during the sessions was 28.1 ± 0.5° and 28.2 ± 0.4°, respectively. No statistical differences were found between static and dynamic sessions at the start and on the average in the environmental temperature ($p > 0.73$).

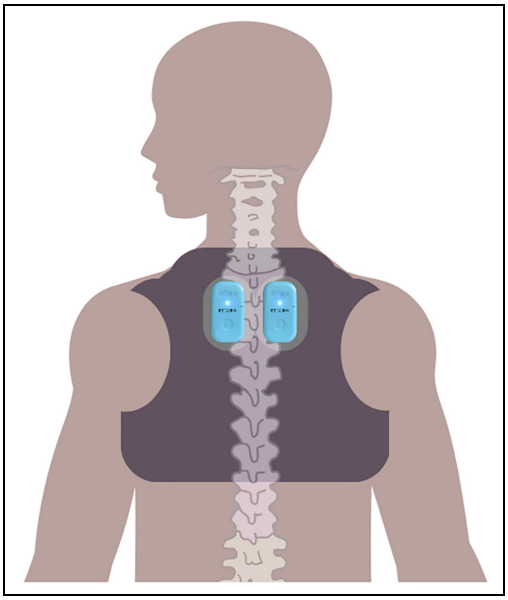

Figure 1. Placement and location of both devices in the players' specific harness during the (dynamic trial).

Statistical analysis

Each IMU's internal temperature, resultant acceleration and PL_{RT} were exported from SPRO™ software to an MS Excel worksheet. A descriptive analysis was performed, and data were shown as the mean and standard deviation of the temperature and average resultant acceleration for the individual 5-min intervals. The Shapiro–Wilk and Levene tests were performed to analyse normality and homoscedasticity, fulfilling parametric inference requirements. Statistical differences between device parameters (a) internal temperature, (b) resultant acceleration and (c) PL_{RT} were analysed by repeated-measures *t*-test. The magnitude of differences was calculated using: (1) percentage of coefficient of variation (CV%)[18]; (2) intraclass correlation coefficient (ICC)[19] interpreted as: 0 *poor*; 0.01–0.20 *trivial*; 0.21–0.40 *regular*; 0.41–0.60 *moderate*; 0.61–0.80 *substantial*; 0.81–0.99 *almost perfect*; 1.00 *perfect*); and (3) Cohen's *d* (*d*) interpreted as: < 0.2 *trivial*; 0.2–0.49 *small*; 0.5–0.79 *moderate* and > 0.8 *large*).[20] Statistical analyses and plots were performed and designed using GraphPad Prism (release 8, GraphPad Software, La Jolla, CA, USA). Statistical differences were considered if $p < 0.05$.

Results

Figure 2 shows the comparative results of the two operating temperatures (environmental conditions vs optimal working temperature; blue vs red colour) at static (2a, 2c and 2e) and dynamic (2b, 2d and 2e) conditions on the IMUs' internal temperature, resultant vector of acceleration and PL_{RT}.

Regarding the internal temperature, at the static conditions (Figure 2(a)) differences were found with high effect size from the start to minute 22 ($p < 0.001$; $d = 0.88$–4.31; CV = 2.37–14.04%; ICC = 0.08–0.24), moderate effect size from minute 23 to 27 ($p < 0.01$; $d = 0.51$–0.79; CV = 1.36–2.15%; ICC = 0.26–0.48), and small effect size from minute 28 to 31 ($p = 0.01$–0.04; $d = 0.28$–0.45; CV = 0.87–1.21; ICC = 0.66–0.81). On the other hand, at the dynamic condition (Figure 2(b)), differences were found with a high effect size from the start to minute 12 ($p < 0.001$; $d = 0.80$–1.54; CV = 5.18–16.99%; ICC = 0.09–0.38), moderate effect size from minute 13 to 16 ($p < 0.01$; $d = 0.51$–0.66; CV = 2.96–4.59%; ICC = 0.29–0.71), and small effect size from minute 17 to 20 ($p = 0.01$–0.04; $d = 0.22$–0.48; CV = 1.16–2.42%; ICC = 0.58–0.67).

In the resultant vector of acceleration, differences were found at the static conditions (Figure 2(c)) with a high effect size from the start to minute 13 ($p < 0.001$; $d = 0.84$–1.97; CV = 1.12–1.88%; ICC = 0.24–0.63), with a moderate effect size from 14 to minute 22 ($p < 0.01$; $d = 0.54$–0.79; CV = 0.72–1.06%; ICC = 0.26–0.42) and with a small effect size from 23 to minute 29 ($p < 0.01$; $d = 0.26$–0.49; CV = 0.48–0.69%; ICC = 0.34–0.73). Secondly, differences were found at the dynamic session with a moderate effect size from the start to minute 2 ($p < 0.01$; $d = 0.51$–0.62; CV = 2.15–2.57%; ICC = 0.95–0.97), and with small effect size from minute 3 to minute 13 ($p < 0.01$; $d = 0.21$–0.40; CV = 0.95–1.82%; ICC = 0.88–0.99).

Finally, the PL_{RT} dynamics and accumulated PL_{RT} are shown in Figure 2(e) and (f) and 2(g) and (h) respectively. No differences between operating temperature conditions were found in PL_{RT} dynamics and accumulated PL_{RT} at static ($p = 0.06$; $t = 2.11$; $d = 0.19$; CV = 4.83%; ICC = 0.90) and dynamic conditions ($p = 0.71$; $t = 0.38$; $d = 0.08$ trivial; CV = 2.62%; ICC = 0.97).

Discussion

Three-axis accelerometers need constant temperature for optimal functioning.[12,13] Previous studies have shown that a change of 1°C relative to 25°C may result in a 0.4–1.2 mg change in acceleration values when using low-cost acceleration sensors.[14,15] Internal temperature information may help improve the quality of the accelerometry-based data.[15] In this sense, some manufacturers used inner temperature values to calibrate accelerometers automatically. Based on the result of the present study, after switching on, it required approximately 20–30 min for the IMU device to reach the constant and optimal operating temperature recommended by the manufacturer during static and dynamic sessions. Once the accelerometer achieves this temperature, the data may record a stable measuring of accelerometer-based variables as resultant acceleration.

A previous study has shown that a gradual increasing and decreasing temperature between 15 and 35°C did not affect accelerometry variables (PlayerLoad)[21]

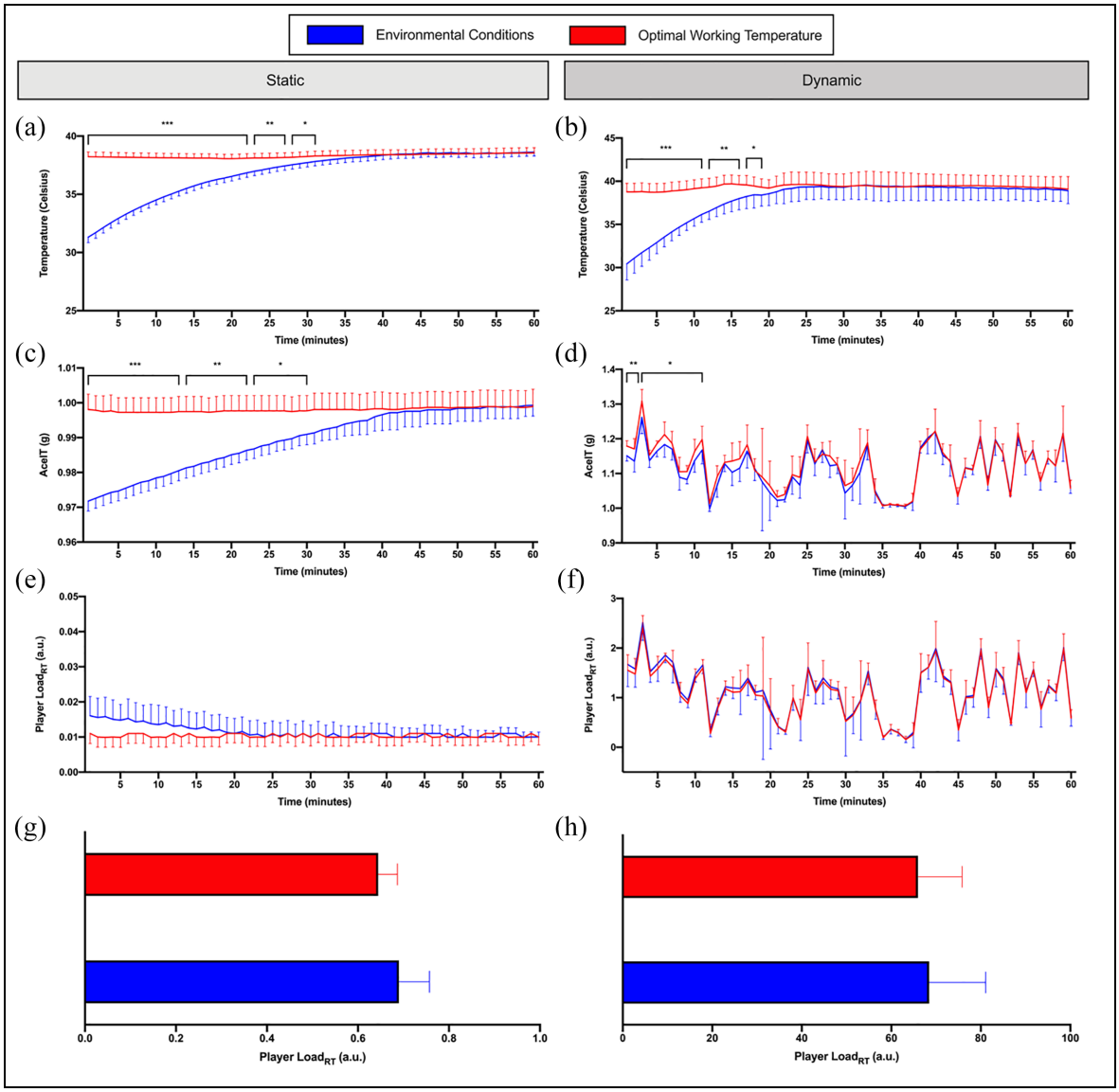

Figure 2. Average, standard deviation and statistical differences between operating temperatures (optimal working temperature, red; vs environmental temperature, blue) values in the internal temperature (Celsius grades) at (a) static and (b) dynamic session; resultant acceleration (*g* force) at (c) static and (d) dynamic session; PL_{RT} (arbitrary units, a.u.) in (e) static and (f) dynamic session in 1-min periods, and accumulated PL_{RT} (a.u.) in the entire session at (g) static and (h) dynamic sessions.
$^{**}p < 0.01$, $d > 0.5$. $^{**}p < 0.05$, $d > 0.2$. $^{***}p < 0.001$, $d > 0.80$.

within a session. These results have been obtained in the present study in PL_{RT} dynamics and the accumulated PL_{RT} of the entire session. PL is an algorithm that sums the differences between resultant vector values throughout the session, in other words, PL is related to the magnitude of changes in acceleration but not the magnitude of acceleration itself.[22] Although the resulting acceleration was affected for the first 30-min, it was used to calculate different variables such as linear velocity, range of movement and mobility during strength training,[23] segment stability[24] or impacts on other parts.[17] Also, there could be differences in accumulated (magnitude) acceleration but with no impact in PL_{RT} dynamics, these differences are due to the way both variables are estimated. The magnitude of the acceleration is derived by the sum of acceleration, but PL_{RT} is the sum of the differences between two successive accelerations.[5]

It should be considered that although there were significant differences in the first 30-min during both dynamic and static trials independently, this internal temperature would not influence accumulated workload under static or dynamic conditions. Coaches, athletes and sports scientists may understand that some factors could affect the outcome of accelerometry-based variables in IMUs.[3,4] As in other instruments used in sports science, stakeholders have to be careful to follow protocols considering different technical aspects such as previous calibration, chipsets, filtering methods, a variety of sampling frequencies and data-processing

algorithms.[5,15] Therefore, it is important to perform a regular calibration of the accelerometers following the manufacturer's guidelines and switch on the devices a substantial time before the sessions to obtain the optimal operating temperature to avoid bias produced by the temperature drift at the start of the sessions.

Limitations

While the results of this study have provided information about the impact of operating temperature on the resultant acceleration vector and PL_{RT}, some limitations to the examination must be acknowledged. Because the company carries out different filtration processes before the data are available for the user, the results should only be specific to the analysed device (e.g., brand and model). Future studies encourage replicating this analysis with raw accelerometer data without company filtration, being data post-processed with transparent filters that enable other researchers or practitioners to repeat the assessment with different devices. Another limitation that needs consideration is the environmental temperature. Although the conditions are similar in the two sessions (no differences in average and start temperature), the results are specific to this temperature, and the time to stabilize will change depending on the environmental temperature. Finally, some studies considering the devices measurement error should be performed to confirm differences are based on temperature only or there is device measurement error. Also, these data variances should be in-depth evaluated considering its practical meaningful when analysing different exercise related tasks.

Conclusions

From the results obtained in the present study, different conclusions can be mentioned:

1. Accelerometer devices should reach the optimal operating temperature when Total Acceleration was used as a reference variable (vector sum of acceleration in the three-axis of motion). Instead, $PlayerLoad^{RT}$ is not affected by temperature because it quantifies the accumulation in the change of Total Acceleration.
2. Under dynamic conditions, the inertial device reaches optimum operating temperature more quickly than under static conditions (20 vs 30 min). In static conditions, the accelerometer only acquires temperature due to its own functioning and from the components that form the inertial device, while in dynamic conditions the transfer of the athlete's body heat is added.

Practical applications

Different recommendations could be provided for sports sciences professionals that use accelerometry-based devices: (a) variables dependent on the resultant vector (e.g. linear velocity in strength training, segments stability, range of movement and mobility, jumps and others), the devices should be used when the optimal operating temperature has been reached; (b) variables dependent of accumulated workload (e.g. PlayerLoad), the environmental devices overestimated the total load although significant differences were not obtained. Therefore, a standardized protocol before every session when using variables based on accelerometry seems necessary to achieve optimal measurements.

Declaration of conflicting interests

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References

1. Cummins C, Orr R, O'Connor H, et al. Global Positioning Systems (GPS) and microtechnology sensors in team sports: a systematic review. *Sports Med* 2013; 43: 1025–1042.
2. Whitehead S, Till K, Weaving D, et al. The use of microtechnology to quantify the peak match demands of the football codes: a systematic review. *Sports Med* 2018; 48: 2549–2575.
3. Camomilla V, Bergamini E, Fantozzi S, et al. Trends supporting the in-field use of wearable inertial sensors for sport performance evaluation: a systematic review. *Sensors* 2018; 18: 873.
4. Gómez-Carmona CD, Bastida-Castillo A, Ibáñez SJ, et al. Accelerometry as a method for external workload monitoring in invasion team sports. A systematic review. *PLoS ONE* 2020; 15: e0236643.
5. Gómez-Carmona CD, Pino-Ortega J, Sánchez-Ureña B, et al. Accelerometry-based external load indicators in sport: too many options, same practical outcome? *Int J Environ Res Public Health* 2019; 16: 5101.
6. Gómez-Carmona C, Rojas-Valverde D, Rico-González M, et al. What is the most suitable sampling frequency to

register accelerometry-based workload? A case study in soccer. *Proc IMechE, Part P: J Sports Engineering and Technology* 2021; 235(2): 114–121.
7. Migueles JH, Cadenas-Sanchez C, Ekelund U, et al. accelerometer data collection and processing criteria to assess physical activity and other outcomes: a systematic review and practical considerations. *Sports Med* 2017; 47: 1821–1845.
8. Gómez-Carmona CD, BASTIDA-Castillo A, García-Rubio J, et al. Static and dynamic reliability of WIMU PRO^{TM} accelerometers according to anatomical placement. *Proc IMechE, Part P: J Sports Engineering and Technology* 2019; 233: 238–248.
9. Nedergaard NJ, Robinson MA, Eusterwiemann E, et al. The relationship between whole-body external loading and body-worn accelerometry during team-sport movements. *Int J Sports Physiol Perform* 2017; 12: 18–26.
10. Edwards S, White S, Humphreys S, et al. Caution using data from triaxial accelerometers housed in player tracking units during running. *J Sports Sci* 2019;37(7): 810–818.
11. Mitschke C, Kiesewetter P and Milani TL. The effect of the accelerometer operating range on biomechanical parameters: stride length, velocity, and peak tibial acceleration during running. *Sensors* 2018; 18: 130.
12. Lee KI, Takao H, Sawada K, et al. Low temperature dependence three-axis accelerometer for high temperature environments with temperature control of SOI piezoresistors. *Sens Actuators Phys* 2003; 104: 53–60.
13. He J, Xie J, He X, et al. Analytical study and compensation for temperature drifts of a bulk silicon MEMS capacitive accelerometer. *Sens Actuators Phys* 2016; 239: 174–184.
14. Niu X, Li Y, Zhang H, et al. Fast thermal calibration of low-grade inertial sensors and inertial measurement units. *Sensors* 2013; 13: 12192–12217.
15. van Hees VT, Fang Z, Langford J, et al. Autocalibration of accelerometer data for free-living physical activity assessment using local gravity and temperature: an evaluation on four continents. *J Appl Physiol* 2014; 117: 738–744.
16. Waegli A, Skaloud J, Guerrier S, et al. Noise reduction and estimation in multiple micro-electro-mechanical inertial systems. *Meas Sci Technol* 2010; 21: 065201.
17. Gomez-Carmona CD, Bastida-Castillo A, González-Custodio A, et al. Using an inertial device (WIMU PRO^{TM}) to quantify neuromuscular load in running: Reliability, convergent validity and the influence of type of surface and device location. *J Strength Cond Res* 2020; 34: 365–373.
18. Atkinson G and Nevill AM. Statistical methods for assessing measurement error (reliability) in variables relevant to sports medicine. *Sports Med* 1998; 26: 217–238.
19. Kramer MS and Feinstein AR. Clinical biostatistics. LIV. The biostatistics of concordance. *Clin Pharmacol Ther* 1981; 29: 111–123.
20. Cohen J. *Statistical power analysis for the behavioral sciences*. 2nd ed. Hillsdale, NJ: Erlbaum, 1988.
21. Boyd LJ, Ball K and Aughey RJ. The reliability of minimaxX accelerometers for measuring physical activity in Australian football. *Int J Sports Physiol Perform* 2011; 6: 311–321.
22. Bredt S, da GT, Chagas MH, Peixoto GH, et al. Understanding player load: meanings and limitations. *J Hum Kinet* 2020; 71: 5–9.
23. Pino-Ortega J, Bastida-Castillo A, Hernández-Belmonte A, et al. Validity of an inertial device for measuring linear and angular velocity in a leg extension exercise. *Proc IMechE, Part P: J Sports Engineering and Technology* 2020; 234: 30–36.
24. Pino-Ortega J, Hernández-Belmonte A, Gómez-Carmona CD, et al. Monopodal postural stability assessment by wireless inertial measurement units through the fast Fourier transform. *J Sport Rehabil* 2019; 29: 738–747.